

5ai	Since the <u>field strength is zero at a point between the spheres</u> / there are <u>two sections of the graph that are of opposite signs</u> hence <u>opposite direction of E-field</u> therefore the charges are the <u>same sign</u>.	M1 A1
5aii	At <u>$x = 0.08 \text{ m}$</u>, the electric field strength due to sphere A cancels out the electric field strength due to sphere B OR net E is zero. $E_A = E_B$ $\frac{Q_A}{4\pi\epsilon_0 (0.080)^2} = \frac{Q_B}{4\pi\epsilon_0 (0.040)^2}$ $\frac{Q_A}{Q_B} = \left(\frac{0.080}{0.040}\right)^2 = 4.0$	M1 M1 A1

5bi	$E = V/d = 100 / 0.050 = 2000 \text{ N C}^{-1}$ Direction of electric field strength is upwards (high to low potential)	B1 B1
5bii	$a = \frac{qE}{m}$ $= ((1.6 \times 10^{-19}) (2000) / (9.11 \times 10^{-31}))$ $= 3.5 \times 10^{14} \text{ m s}^{-2}$	C1 A1
5biii	$\Delta U = q\Delta V$ $= (-1.6 \times 10^{-19}) [(-100) - (-150)]$ $= -8.0 \times 10^{-18} \text{ J}$	C1 A1

6a	change in magnetic flux density $\Delta B = 5 - 50 = -45 \text{ mT}$ (from graph, accept positive)	B1
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